19

Interstellar absorption

19.1 Introduction

Even though this volume is dedicated only to stellar observations and the theory of stellar structure, we have to talk a little about the interstellar material, which means the material which is between the stars, because the light of the stars, especially of those which are very far away, has to pass through this interstellar material before it reaches us. Just as in the Earth's atmosphere, the star light is influenced by interstellar absorption, also called interstellar extinction. We cannot observe all the light which is emitted from the star because the interstellar gas and the interstellar grains, called dust, have absorbed part of the light. Just as for the Earth's atmosphere, we have to correct for interstellar absorption. For the Earth's atmosphere we can determine the extinction if we observe the star at different zenith distances, which means for different path lengths, through the atmosphere. For interstellar extinction we cannot do this since the path length through the interstellar medium is always the same. We have to find other ways to determine the influence of the interstellar extinction.

Basically there are two components of interstellar extinction: the absorption by the grains and the absorption by the gas. These components are usually, but not necessarily, associated. In the following paragraphs, we will discuss both components and see how we can find out about interstellar absorption.

19.2 The interstellar dust

19.2.1 General appearance

When looking at photographs of the Milky Way, we see regions with a large number of stars and, right next to them, regions where there are hardly any stars.

Dark patches intermixed with bright matter can be seen in many parts of the sky (see, for instance, Fig. 19.1). On the small-scale photograph of the Milky Way seen in Fig. 19.2 a dark lane cuts the Milky Way almost into two halves.

The question is, are the stars distributed so inhomogeneously, or is there something covering up the stars? When we try to find external galaxies, we

Fig. 19.1. In the sky, dark regions with very few stars are seen next to brighter regions with very many stars. In this picture we see the so-called Horsehead Nebula. (From Abell 1982.)

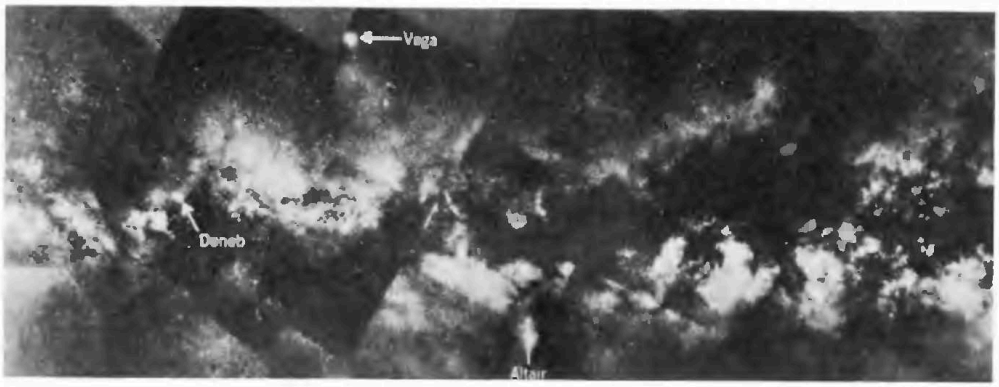

Fig. 19.2. A composite photograph of our Milky Way is shown. The border lines for different photographic plates can clearly be seen. We also see large dark lanes cutting through the Milky Way almost cutting it in two halves. (From Shu 1982.)

have trouble finding any in the plane of the Milky Way, which means in the plane of our Galaxy. For this reason the plane of the Galaxy, or of the Milky Way, is called the zone of avoidance. This strongly suggests that there is indeed absorption in the plane of the Galaxy because the external galaxies have no reason to care about the plane of our Galaxy. How can we decide with certainty whether the dark regions are indeed regions of strong absorption and not holes in the Milky Way? We can find out by statistics, by means of the so-called Wolf diagram.

19.2.2 The Wolf diagram

Let us look at a certain region in the sky and thereby observe all the stars in a given solid cone with an opening $\Delta\omega$. All these stars are projected against the given area in the sky (see Fig. 19.3). We study only stars of a given kind, say only A0V stars or, in other words, stars with spectra, like Vega. All such stars have the same absolute magnitude, M_v. (There are more luminous A0 stars around like giants and supergiants, but we saw that they can be recognized from their spectra.)

Let us assume that the stars are distributed homogeneously in this cone and that the number of A0V stars per pc^3 is n. In any volume V, the number of A0V stars is then $N = n \cdot V$. We now consider the part of the spherical shell at distance d with thickness Δd, which is cut out by the cone (see Fig. 19.3). The whole sphere is described by a solid angle 4π. The volume of the part of the sphere with solid angle opening $\Delta\omega$ is therefore given by a fraction $\Delta\omega/4\pi$, which means

$$V = 4\pi d^2 \, \Delta d \, \Delta\omega/4\pi. \tag{19.1}$$

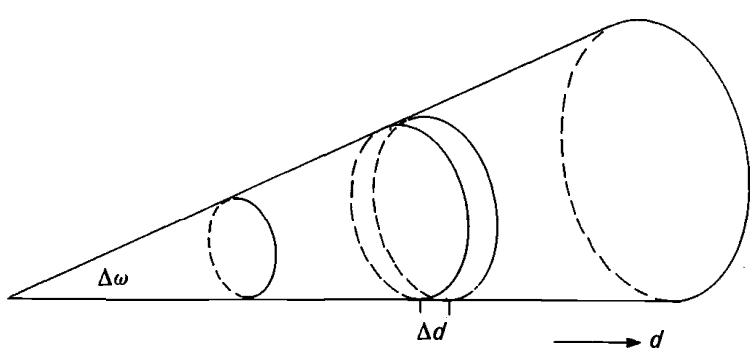

Fig. 19.3. Observing all the stars in a given area in the sky corresponds to observing all stars within a cone with an opening of solid angle $\Delta\omega$, whose size is determined by the extent of the observed region in the sky.

The number of A0V stars in this volume is then

$$N(d) = n \cdot V = n \cdot d^3 \cdot \Delta\omega \cdot \frac{\Delta d}{d}. \tag{19.2}$$

These stars all have an apparent visual magnitude

$$m_v = M_v + 5 \log d - 5, \tag{19.3}$$

or, for a change in distance by $\Delta \log d$ we find a change in apparent visual magnitude by

$$\Delta m_V = 5 \cdot \Delta \log d = 5 \cdot \log e \cdot \Delta \ln d = 2.171 \cdot \Delta \ln d = 2.171 \cdot \Delta d/d. \tag{19.4}$$

(A given Δm_V corresponds to a given $\Delta \ln d = \Delta d/d = \Delta m_V/2.171$.) For a given m_V we have then, according to (19.3)

$$\log d = \frac{m_v - M_v + 5}{5} = \frac{m_v}{5} - \frac{M_v}{5} + 1, \tag{19.5}$$

where d is obtained in pc. M_v is known from nearby A0V stars like Vega; m_v is measured.

The number of A0V stars with apparent magnitudes between m_V and $m_V + \Delta m_V$ is given by the number of stars with distances between d and $d + \Delta d$, for which the relation between Δm_V and $\Delta d/d$ is obtained from (19.4). From (19.2) we find

$$\log N[m_v(d)] = \log n + 3 \cdot \log d + \log(\Delta\omega \cdot \Delta \ln d), \tag{19.6}$$

if we assume a homogeneous stellar density with n stars per unit volume, which means $n =$ constant. For a given choice of $\Delta \ln d$ corresponding to a chosen Δm_V for the magnitude interval for which we count the stars, all terms on the right-hand side of (19.6) are constant except for the $\log d$ term, which depends on the m_V for which we count the stars. We can therefore rewrite (19.6) in the form

$$\log N[m_v(d)] = 3 \cdot \log d + \text{const}, \tag{19.7}$$

with

$$\text{const} = \log \Delta\omega + \log \frac{\Delta m_v}{2.171} + \log n.$$

Replacement of $\log d$ by means of (19.5) now yields

$$\log N[m_v(d)] = 0.6\, m_v + \text{Const}, \tag{19.7a}$$

where the new Constant is different from the constant in (19.7) because it also includes the term stemming from the absolute magnitudes of the stars studied.

In a plot of $\log N$ versus m_V we should therefore find a straight line (see Fig. 19.4). This is the Wolf diagram.

Assume now that there is an absorbing cloud at distance $d = d_0$, for which $m_V = m_{V0}$. Suppose this cloud absorbs 75% of the light of the stars behind it, leaving 25% of the light, which means decreasing the light by a factor of 4, corresponding to an increase in magnitude by 1.5 for all stars behind the cloud. For a given distance, of course, the number of stars would not change; the $N(d)$ for $d > d_0$ would remain the same, but the apparent magnitude of these stars would be increased by 1.5. The relation between $\log N$ and m_v would then be as shown by the dashed line in Fig. 19.4. For the figure we also assumed that the cloud would extend over a distance Δd, corresponding to $\Delta m_v = m_{v1} - m_{v0}$. Without this finite extent to the cloud, the short, dashed line in the figure would have to be plotted horizontally because the m_v of the stars would either be changed by 1.5 or not at all. If the cloud is extended, the stars in the cloud have intermediate shifts, depending on their position in the cloud.

If at a larger distance $d = d_1$ there should be another cloud, we may find another shift in the distribution, as indicated by the dash–dotted curve in Fig. 19.4. Such a diagram can therefore not only tell us whether there is a cloud or not, but also where it is, how extended it is, and how much it absorbs.

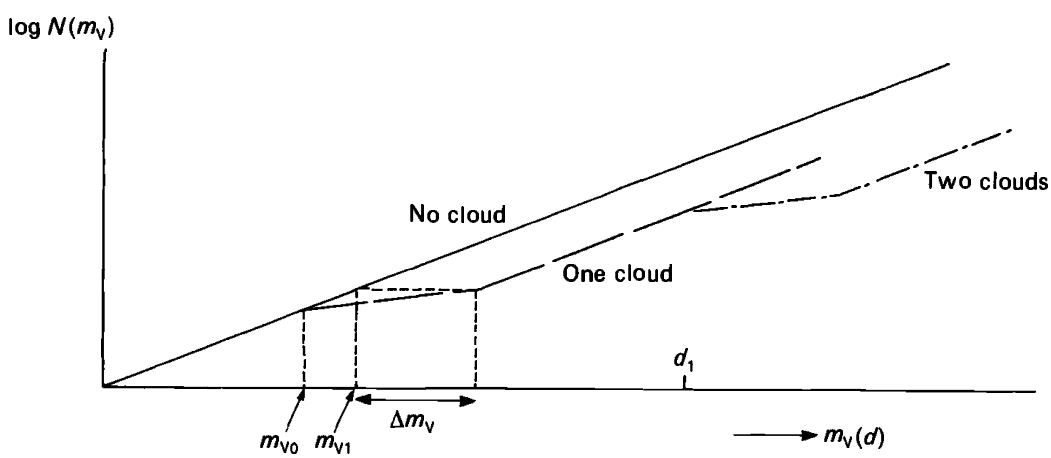

Fig. 19.4. A schematic Wolf diagram is shown. For homogeneous steller densities we expect a linear relation between the log of the number of stars of a given kind, log N (with a given absolute visual magnitude) and the apparent visual magnitudes (see the solid line). If there is an absorbing intersteller cloud between us and the stars the apparent magnitudes of the stars behind the cloud are all increased by an amount determined by the absorption in the cloud. The relation between $\log N$ and m_v is shifted as shown by the long dashed line. A second cloud behind the first one may give an additional shift (dash–dotted line).

What would we see if there was not an absorbing cloud, but instead there was a 'hole' in the star distribution, say, between the distances d_2 and d_3? If these distances correspond to apparent magnitudes m_{v2} and m_{v3}, then all the stars with $m_{v2} < m_v < m_{v3}$ would be missing. For these magnitudes N would be zero. The stars at larger distances would generally not be affected, but would probably have the same average density. The star counts for larger m_v would therefore be the same. The relation between $\log N$ and m_v would then look like the one shown in Fig. 19.5, where we have assumed that two holes exist: one at $m_v = m_{v2}$ and another one at $m_v = m_{v4}$. In this case the Wolf diagram looks very different from the one with the clouds. Such a Wolf diagram therefore permits us to distinguish between 'holes' in the stellar distribution and an absorbing interstellar cloud. In Fig. 19.6 we show an example of an observed diagram, clearly showing that interstellar clouds are obscuring the stars.

19.2.3 Interstellar reddening

The study described in the preceding section can, of course, also be done with the blue and ultraviolet magnitudes in the same way. We can then find out how much the same clouds absorb in the blue or in the violet. Such studies have shown that the absorption is larger in the blue and still larger in the ultraviolet. For most clouds we find

$$\Delta m_B = 1.3 \cdot \Delta m_v \quad \text{and} \quad \Delta m_U = 1.53\, \Delta m_v. \tag{19.8}$$

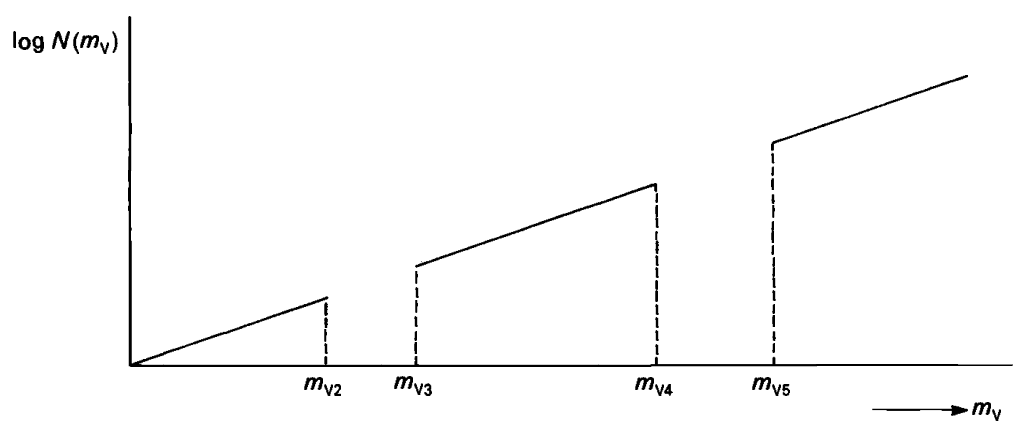

Fig. 19.5. We show the Wolf diagram expected if there are holes in the number density of stars at distances corresponding to apparent visual magnitudes m_{V2} and m_{V4}, but which would be normal again for distances corresponding to apparent visual magnitude m_{V3} and m_{V5}, respectively. In this case we would see no stars with apparent magnitudes between m_{V2} and m_{V3} or with magnitudes between m_{V4} and m_{V5}. There would be 'holes' in the $\log N(m_V)$ plot.

Since the B and U magnitudes are more increased than the V magnitudes, the colors of the stars change due to the interstellar absorption. We find, roughly,

$$\Delta m_B - \Delta m_v = \Delta(B - V) = \Delta m_v(1.31 - 1) = \Delta m_v \cdot 0.31, \tag{19.9}$$

or

$$\Delta m_v = 3.2 \cdot \Delta(B - V) \tag{19.10}$$

or

$$\frac{\Delta(B - V)}{\Delta m_v} = 0.31 \quad \text{and} \quad \frac{\Delta(U - B)}{\Delta m_v} = 0.22 \tag{19.11}$$

$$\Delta(U - B)/\Delta(B - V) = 0.72, \tag{19.12}$$

but small differences may exist between different clouds.

In order to find the relation between the color change $\Delta(B - V)$ or $\Delta(U - B)$, we can, of course, also directly compare the colors of the A0V stars

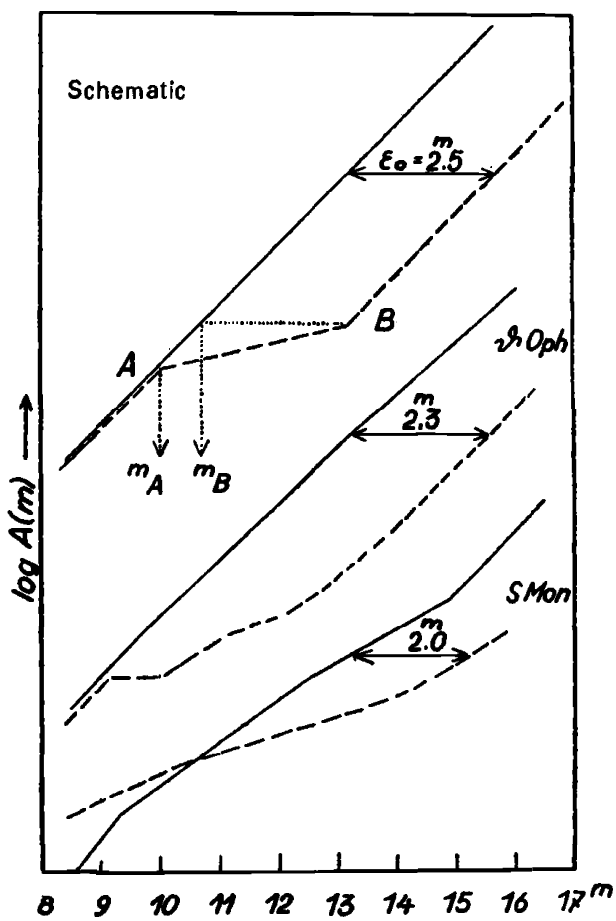

Fig. 19.6. Below the schematic Wolf diagram, already shown in Fig. 19.4, we see actually observed Wolf diagrams for a region around the star ϑ Ophiuchi and around the star S Monocerotis. The stars behind the cloud are dimmer by 2.3 and 2.0 magnitudes due to the absorption by the interstellar material in the clouds. (From Becker 1950).

in front of the cloud with those behind the cloud and determine the color changes and the ratio of $\Delta(B-V)$ to $\Delta(U-B)$.

In the far infrared the interstellar absorption becomes very small, actually close to zero. If we call a far infrared magnitude m_{in}, then the color change $\Delta(m_V - m_{in})$ for stars behind the cloud gives directly the change in visual magnitude for the stars behind the cloud. In this way the ratio of Δm_V to $\Delta(B-V)$, called R, is determined. It usually turns out to be $R = 3.2$, but may occasionally have values around 5 or perhaps even 6.

The $\Delta(B-V)$ and $\Delta(U-B)$ due to interstellar absorption are usually called $E(B-V)$ and $E(U-B)$, where E stands for color excess.

Once we have established that we do indeed have absorption in the interstellar medium, and once we know that the interstellar medium causes reddening, we can use this reddening effect to correct for the interstellar absorption. If we know what the true $U-B$ and $B-V$ colors are for unreddened stars, we can plot the two-color diagram for these true colors (see Fig. 19.7). In this figure we have also plotted the observed colors for some stars. For a given $U-B$ the reddened stars have a much larger $B-V$. We know now that for any change in $B-V$ by a value $E(B-V)$ the change in $U-B$ is $E(U-B)$ with $E(U-B) = 0.72 \times E(B-V)$. This relation defines a direction in the two-color plot. In Fig. 19.7 we have indicated the direction in which the stellar colors move in the two-color diagram due to interstellar reddening. For an $E(B-V) = 0.3$, star D would appear in the position of star C. We can then trace backwards from where the star in position C came in the two-color diagram before it was reddened by tracing the observed colors

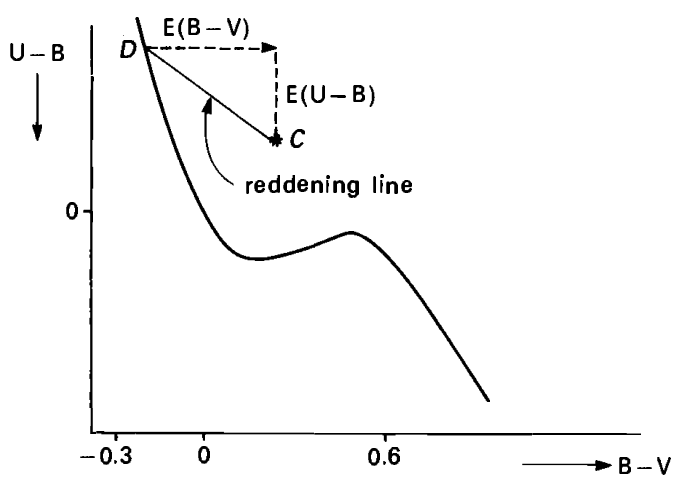

Fig. 19.7. The two-color diagram for the true colors of the stars is shown schematically by the solid line. Also shown is a point (*) for the observed colors of a star C. The direction of the reddening line $E(U-B)/E(B-V) = 0.72$ is also drawn in. Following the direction to the 'true' color line of normal stars tells us what the unreddened colors are of the observed star (point D).

back along the reddening line to the true color line. This gives us the true colors of the star. Since $\Delta m_v = 3.2 \times E(B-V)$, we can also derive by how much the apparent magnitude of the star has been changed due to the interstellar absorption and can correct for it. Suppose we have measured for an A0V star a $B-V = 0.10$ and $U-B = 0.07$. Its true $B-V = 0$ as we know for an A0V star. Its $E(B-V)$ is therefore $E(B-V) = 0.10$ and therefore $\Delta m_v = 0.32$. If we measured $m_v = 8.32$, its true apparent magnitude would have been $m_{v0} = 8.0$ if there had not been any interstellar absorption. If we want to determine the distance to the star we can use (19.3), but we have to use, of course, the apparent magnitude which the star would have had without the interstellar absorption, i.e. m_{v0}. With $M_V = 0.5$ we find a distance modulus $m_v - M_V = 7.5$, which puts the star at a distance of about 316 pc.

On average, we find an interstellar absorption of about one magnitude in the visual for a distance of 1 kpc. The distance to the galactic center is about 8 kpc. Even without the dense dark clouds in that direction, the stars close to the center would be expected to have an average reduction in brightness of 8 magnitudes due to the average interstellar extinction. Their colors are expected to be changed by $E(B-V) = 2.4$.

19.2.4 *Determination of the 'true' colors of the stars*

How can we determine the 'true' color line for the stars? We can, for instance, measure the colors of nearby stars. Chances are that they are nearly unreddened. There are, however, no O or B stars nearby. If there are no nearby stars of a given kind, the best thing is usually to measure stars which are above the galactic plane, if possible. Stars in the plane are usually more reddened than those above the galactic plane because the dust is confined to a rather thin disk (see Fig. 19.8). In fact, it is this observation which tells us

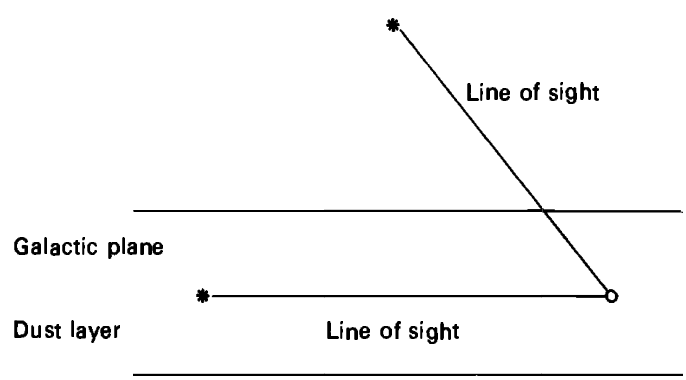

Fig. 19.8. For a given distance stars in the Galactic plane are more reddened than stars outside of the Galactic plane because the line of sight passes through the dust layer for a longer distance.

that the dust is confined to a rather thin disk. It is distributed inhomogeneously, but rarely reaches higher than about 100 pc.

Unfortunately, the O and B stars are also strongly concentrated toward the galactic plane, since they are formed from large dust clouds. So they lie entirely within the dust layer. But the dust layer is patchy and for some stars the E(B − V) is smaller than for others, so we can tentatively assume that the bluest ones have the 'true' color. For some stars there is a way to check this. Suppose we have observed a few nearby A or F stars so that we know their true colors. If the O and B stars happen to be near A or F stars, for instance, in the clusters, we can then measure the reddening for the A and F stars and hope that they will be the same for the hot O and B stars. In young clusters the extinction is, however, patchy and may be different for different stars, as, for instance, for the Pleiades, in which we can see patches of dust, seen as bright nebulosity (see Fig. 19.9), which is very inhomogeneous. The best way would be to measure colors for binaries with one O or B star and the other component being an A or F star. Unfortunately, there are not many binaries with O and B type components around for which the companion colors can also be measured, mainly because O and B stars are so short-lived, and also

Fig. 19.9. On a long-exposure photograph of the Pleiades star cluster patches of nebulosity show up in front of the stars, due to intersteller dust. This interstellar dust appears bright because it reflects the starlight in our direction.

because they are so bright that A or F stars can only be measured if they are well separated from the hotter star. We may sometimes find stars behind the O and B stars for which we may expect at least as much reddening as for the early type star. We then get an upper limit for the reddening. By putting all the available data together we can get a rather good value for the 'true' colors.

19.2.5 *The wavelength dependence of the interstellar extinction*

For nearby unreddened stars the whole energy distribution can be measured, for instance, by so-called scanner observations which measure the intensities in wavelength bands of 10 or 20 Å. If scanner observations are then also made for the same kind of star at a known distance, the flux ratio of the two stars, corrected for the difference in distance, tells us the interstellar extinction law, because

$$\frac{f_2(\lambda)}{f_1(\lambda)} = e^{-\tau_\lambda(\text{interstellar})}. \tag{19.13}$$

Here τ_λ (interstellar) is the optical depth of the interstellar medium between us and the star (see Fig. 19.10). $f_1(\lambda)$ is the flux of the unreddened star, and $f_2(\lambda)$ the flux of the reddened but otherwise similar star after correction for the difference in distance.

With

$$\Delta m_\lambda = -2.5 \cdot \Delta \log f_\lambda, \tag{19.14}$$

we find

$$\begin{aligned} m_\lambda(2) - m_\lambda(1) &= -2.5[\log f_2(\lambda) - \log f_1(\lambda)] \\ &= +2.5 \log e \cdot [\tau_\lambda(2) - \tau_\lambda(1)] = 2.5 \cdot \log e \cdot \tau_\lambda(2) \end{aligned} \tag{19.15}$$

because star 1 is unreddened, so $\tau_\lambda(1) = 0$.

We then obtain $\tau_\lambda(\lambda)$ for every wavelength and can measure how the interstellar extinction varies with the wavelength λ. For the wavelength band

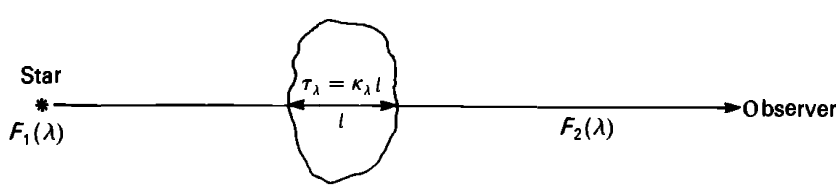

Fig. 19.10. The radiative flux $F(\lambda)$ from a star passes through an interstellar gas cloud and is reduced by the absorption in the cloud. The observed flux $F_2(\lambda) = F_1(\lambda) \cdot e^{-\tau_\lambda}$, where $F_1(\lambda)$ is the flux we would have received without the absorption in the cloud.

$3900 < \lambda < 10\,000$ Å, the wavelength region observed with most telescopes, we find nearly $\tau_\lambda \propto 1/\lambda$. For the atomic absorption of hydrogen and similar atoms or ions we find $\kappa_\lambda \propto \lambda^3$ for the absorption from each atomic level. The interstellar absorption has a very different wavelength dependence. The reason is that the interstellar absorption is mainly due to dust *grains*. The light gets scattered by these little particles. If the particles have a size comparable to the wavelength of the light, the light is not completely blocked out by the grains, but can get around the grain, although not the full amount.

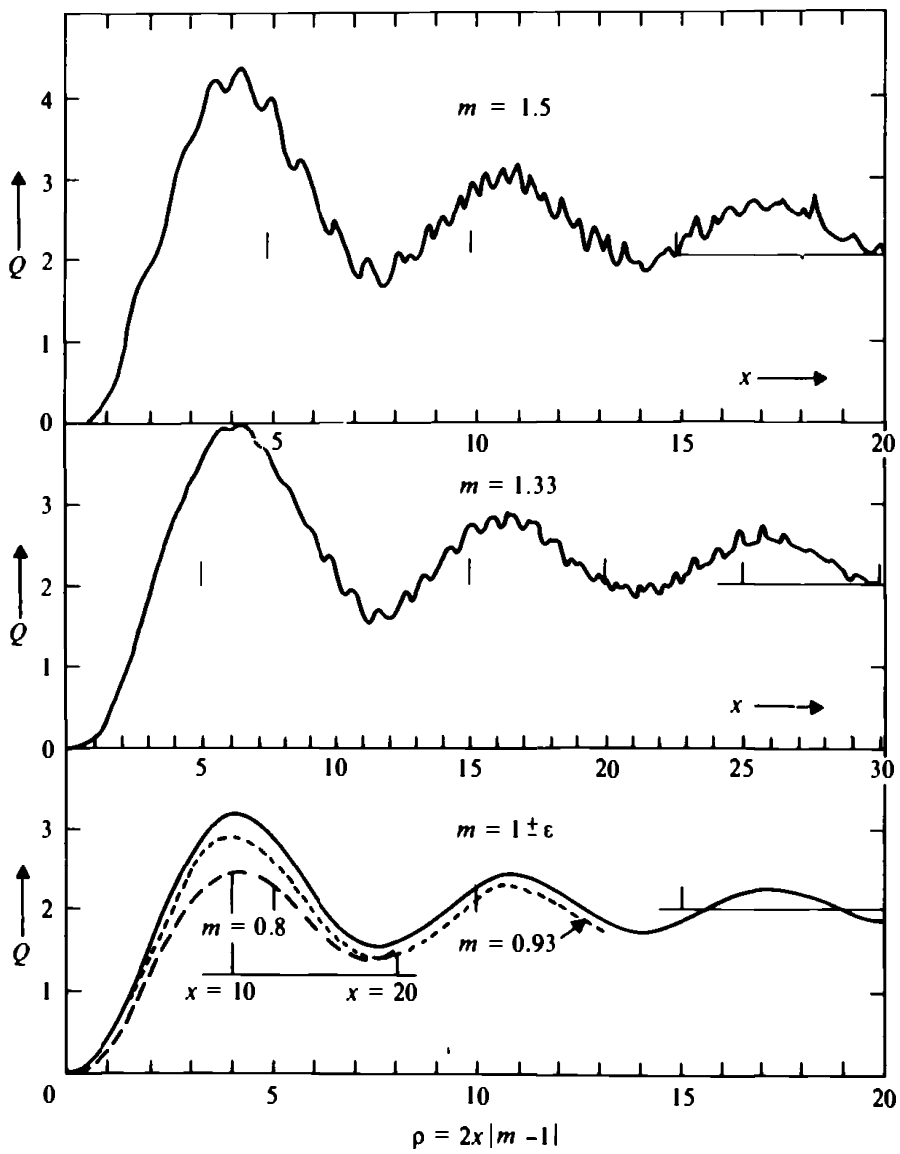

Fig. 19.11. Extinction curves for spherical grains are shown according to Mie's theory. The normalized scattering cross-section $Q = \sigma/(\pi a^2)$ is plotted as a function of $\rho = 2x|m-1|$, where a is the radius of the dust particle, σ the actual cross-section, and x measures the ratio of the particle size to the wavelength. m is the refractive index. (From Dyson & Williams 1980.)

Think about the sound waves with wavelengths of meters which can get around the corner of a house, or the radio waves (100 m) which can also get around houses.

If one calculates the scattering on little spheres, as has been done by Mie, one finds a scattering coefficient, as seen in Fig. 19.1, where the scattering and absorption cross-section is plotted as a function of $\rho = 2x|m-1|$, where

$$x = \frac{2\pi a}{\lambda},$$

and a is the radius of the scattering particles; m is the refractive index; $|m-1|$ is of the order 0.1–0.5 for different substances. This then yields ρ to be about

$$\rho \sim \frac{a}{\lambda}. \tag{19.16}$$

The actual scattering and absorption cross-section for the particles is

$$\sigma = \pi a^2 \cdot Q, \tag{19.17}$$

such that Q gives the ratio of the scattering and absorption cross-section to the geometrical cross-section.

From Figure 19.11 we see that for $1 \leqslant \rho \leqslant 4$ the Q is approximately proportional to $\rho \sim a/\lambda$, or in other words, for $\lambda < a < 4\lambda$, or for dust particle sizes of the order of a few wavelengths of the light the extinction cross-section is roughly proportional to $1/\lambda$. If $a \approx 10^4$ Å, we expect a linear increase of the absorption cross-section σ with $1/\lambda$ for wavelengths between 10 000 and 2500 Å. When ρ becomes very small, which means for very large λ, the extinction goes to zero.

For very long wavelengths the interstellar grains do not absorb or scatter, which means that the interstellar dust clouds become transparent to radio waves. (If you put a football into the path of a sound wave, it does not influence the sound wave.) Therefore, for radio waves the Milky Way dust is transparent. Already in the infrared the extinction becomes very small, almost zero. For very short wavelengths the cross-section for absorption and scattering does not increase any more but remains, finally, about constant with wavelength. Grains much larger than the wavelength of the light simply block the light which hits them. Only the light beams passing between the grains get to us. This fraction of the light is the same for all wavelengths.

19.2.6 The ultraviolet extinction

Since we have rockets and satellites, we can also measure the extinction curves in the ultraviolet. Fig. 19.12 shows the approximate

dependence on $1/\lambda$: the right-hand end of the curve corresponds to the wavelength of 1215 Å, which is the wavelength of the hydrogen Lyα line, which is the strongest hydrogen line because it corresponds to the transition of the electron from the ground level to the next higher level, namely the level with the main quantum number $n = 2$. The interstellar medium absorbs strongly in all the hydrogen Lyman lines originating from the ground level. Therefore, we cannot measure the dust extinction in the wavelength region of these lines. The overall run of the extinction curve in the ultraviolet does not look like that of the Mie curve seen in Fig. 19.11. There is a hump around 2200 Å, which is about 400 Å wide. For shorter wavelengths we find first a decrease and then again a rather steep increase of the absorption.

This behavior may be understood if we have a mixture of particles of different sizes. The smaller particles then become important for shorter wavelengths. The hump needs a special interpretation. Most astronomers believe it is due to very fine graphite particles, although this is not yet certain.

When interpreting these extinction curves, we also have to keep in mind that the Mie theory considers only spherical dust grains. We do not know how the cross-sections change for odd particle shapes.

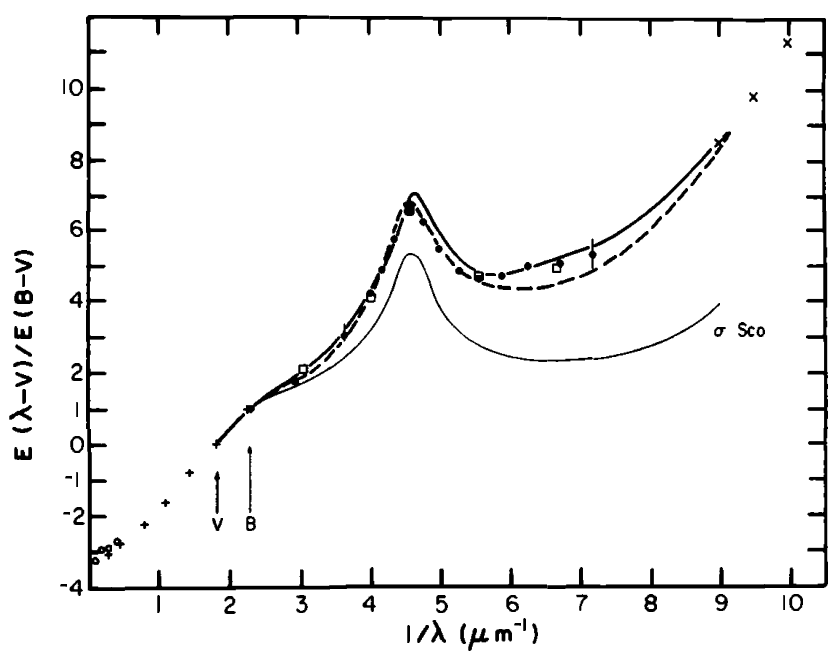

Fig. 19.12. Interstellar extinction curves, $m_\lambda - m_V$, per $E(B-V) = 1$, also written as $(A_\lambda - A_V)/E(B-V)$, or $E(\lambda - V)/E(B-V)$, are shown as a function of $1/\lambda$, where λ is the wavelength of the light measured in μ. ($1\mu = 10^4$ Å). The bump occurs at 2200 Å. The average extinction curve is shown as the dashed line. Low extinctions in the ultraviolet are relatively rare and do not influence the average very much.

The interstellar extinction for the short wavelengths does not look like Mie curves seen in Fig. 9.10. (From Dyson & Williams 1980.)

The interesting part is that this extinction curve is not the same everywhere in the sky. In regions of star formation it looks different. What changes is the size of the hump relative to the rest, and also the degree of increase for very short λ. In dense interstellar clouds, where star formation is going on, the increase at short wavelengths is much smaller and often the hump is less high. In Fig. 19.12 we compare some of these curves with the standard curve. People try to understand this by taking into account the strong ultraviolet radiation of the newly formed hot stars. They speculate that this radiation destroys the small grains, resulting in a smaller increase of the extinction for the short wavelengths.

In the external galaxy next to us, the Large Magellanic Cloud (LMC), the increase towards the ultraviolet is even steeper than shown in Fig. 19.12. The absorbing particles responsible for this increase must be relatively more abundant in the LMC than in the Galaxy. If these were the small grains we wonder why the many young, hot stars in the LMC do not destroy these small grains.

It seems that the interstellar medium in the LMC is, on average, very transparent. There seem to be very few large grains in the LMC.

19.3 The interstellar gas

The interstellar gas is of little importance for the observations of cool stars, except for the observations of the transition layer and coronal lines and the X-ray observations.

For the hot stars it is important that the hydrogen in the interstellar gas absorbs all light being emitted at wavelengths shorter than 912 Å, except for the very closest hot stars, which are, however, all white dwarfs.

Fortunately, cool atomic hydrogen gas can absorb at all wavelengths, which means continuously over all wavelengths, only for $\lambda \leqslant 912$ Å. For wavelengths shorter than 912 Å all hydrogen atoms in the ground level can contribute to this absorption. At 912 Å the absorption coefficient per hydrogen atom is about $10^{-17}\,\mathrm{cm}^2$. With an average density of about 1 hydrogen atom per cm^3 in the interstellar medium (with laboratory standards this is, of course, an extremely good vacuum), there are about 3×10^{19} atoms between us and a star at 10 pc distance. The optical depth τ_λ along the line of sight at 912 Å is then $\tau_\lambda \approx 300$, and the amount of radiation which we would receive is

$$f_\lambda = f_{\lambda 0}\,\mathrm{e}^{-300},$$

where $f_{\lambda 0}$ is the flux which we would have received without any hydrogen absorption. This means we would not be able to measure anything.

Fortunately, there are some directions in the sky for which the hydrogen density along the line of sight is only about 10^{-3} atoms cm^{-3}. In this direction we may then still see far ultraviolet light from hot stars within a distance of about 50 pc. There are, however, no O stars and very few early B stars in this direction within 50 pc, but with an appropriate camera on a satellite those few stars could be observed at these short wavelengths, which would be very important to do.

Our estimates show that generally we cannot expect to receive any of the far ultraviolet light of the O and B stars, unless we go to extremely short wavelengths, namely the X-ray region, where the absorption coefficient of hydrogen is much smaller.

For the O stars most of the radiation is, however, emitted at the unobservable wavelengths. Therefore, we can only observe a relatively small fraction of the total light emitted by these stars. It is therefore very difficult to determine accurate effective temperatures for these stars when we see only a minor fraction of the emitted energy.

The interstellar gas is also quite important for the interpretation of X-ray observations. At X-ray wavelengths the hydrogen and helium absorption coefficients are much lower than the absorption coefficient, around 900 Å, but they are still large enough to absorb a large fraction of the X-rays emitted by hot stars and by extragalactic sources. In Fig. 19.13 we show the calculated dependence of the absorption coefficient on wavelength for the short wavelengths, including the extreme ultraviolet and the X-ray region.

In the optical spectral region the interstellar gas is seen by means of its interstellar absorption lines. Lines of neutral particles, of ionized particles, and of interstellar molecules are seen. In the interstellar medium collisions are very rare. The radiation density usually is also quite low. There are, therefore, very few processes to excite the atoms or ions to higher states of excitation. The interstellar lines therefore usually are due to absorption processes from the ground state of the atoms or ions. Only in very dense regions may we see lines originating from states with very low but non-zero excitation energies. Interstellar absorption lines can usually be recognized easily because only absorption lines from the ground state are seen. In addition, the interstellar lines in the optical region are generally quite sharp. For rapidly rotating hot stars, they can easily be recognized because of this. In Fig. 19.14 we show several spectra in which the interstellar absorption lines from the gas can be seen as sharp lines, while the stellar absorption lines are broadened by rotation.

For these interstellar absorption lines we can also measure the line strengths from which we can determine the number of absorbing atoms

along the line of sight. If we do that for all the lines which can be observed in the optical and in the ultraviolet spectral region, we can determine the relative abundances of the elements along the line of sight. The elements are mostly singly ionized. (Once an ultraviolet photon ionizes an atom, the remaining ion finds it hard to meet another electron with which to recombine because the densities are so low in the interstellar medium.) If we want to study relative abundances, we usually have to look at the lines of the singly ionized ions. There are exceptions, though, for atoms which have rather large ionization energy, like nitrogen or oxygen, and which would need far ultraviolet photons for ionization. These are mainly neutral. Such abundance analysis and the comparison with the abundance of hydrogen

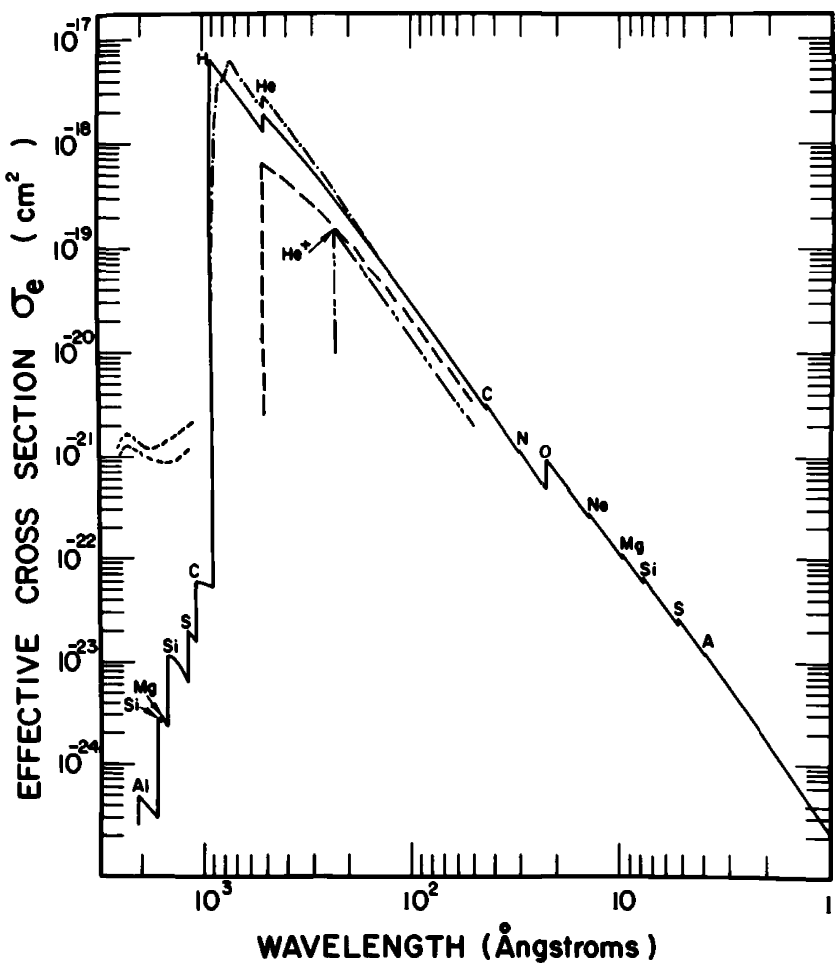

Fig. 19.13. The calculated interstellar extinction is shown also for wavelengths between 100 and 912 Å, which generally cannot be observed. For wavelengths shorter than about 20 Å the extinction becomes smaller than the dust extinction in the visual spectral region. Therefore, X-rays can penetrate the interstellar medium, while radiation with wavelengths between about 100 and 900 Å generally cannot. (From Cruddace *et al.* 1974.)

showed that most of the heavy elements are depleted in the interstellar medium with respect to hydrogen as compared to the solar material.

Obviously, most of the heavy elements are locked up in grains. Clearly, grains cannot form from hydrogen gas. Only a small amount of mass in the form of grains can lock up all the heavy atoms because there are so few of them to begin with. If we want to determine the number of hydrogen atoms absorbing at λ 912 Å from the observed abundance of, say, Ca^+ along the line of sight using the abundance ratio of Ca to hydrogen, we have to take this into account.

For all stellar observations we have to be aware of the effects of interstellar absorption, which may vary considerably from one place to another.

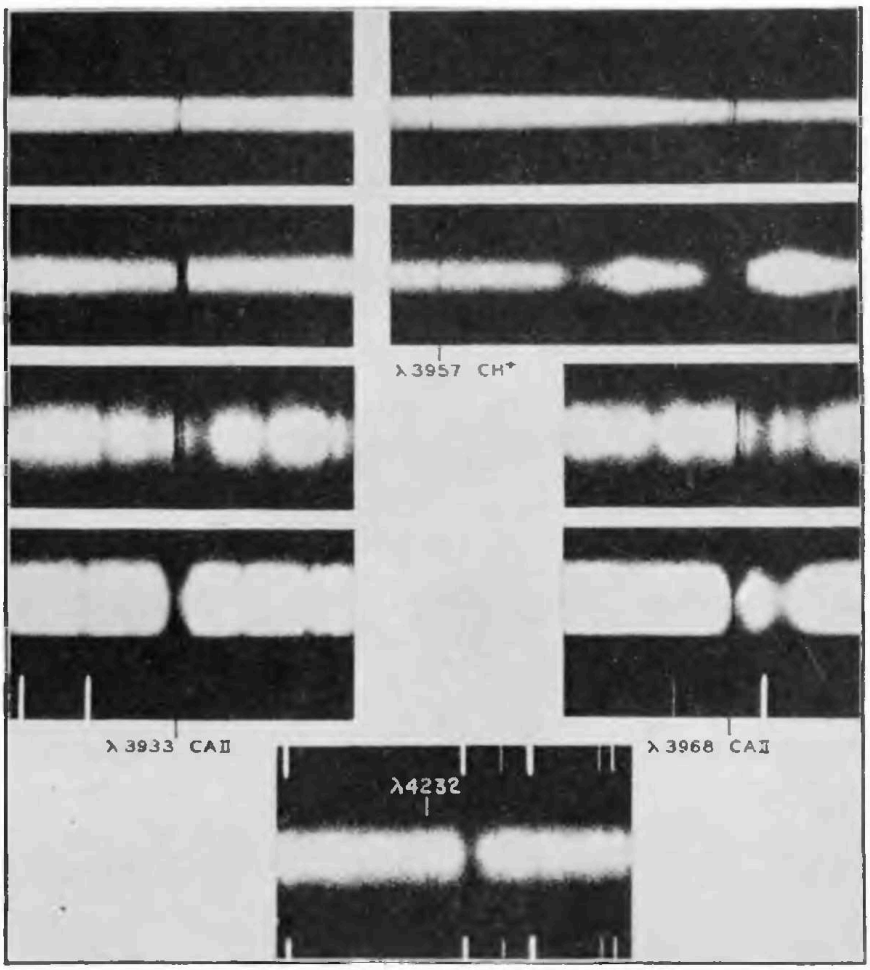

Fig. 19.14. Sharp absorption lines due to absorption by Ca^+ ions in the interstellar gas are seen in the spectra of the stars ζ Ophiuchi (top), P Cygni, υ Saggitarii and α Cygni. In the spectrum of υ Sagittarii a sharp line due to the ionized molecule CH^+ (present also in the interstellar medium) is also seen. (From Greenstein 1951.)